

Determination of Concentration of Acetic Acid in Vinegar.

Vinegar is a sour-tasting liquid produced through the fermentation of ethanol by acetic acid bacteria. It has been used for thousands of years, both as a culinary ingredient and a household staple. Made primarily from fermenting alcohol-based liquids like wine, cider, or grain alcohol, vinegar typically contains 4–8% acetic acid, which gives it its distinct tangy flavor and preservative properties.

In cooking, vinegar enhances flavor profiles, tenderizes meat, and serves as a base for sauces, marinades, and salad dressings. Varieties like balsamic, apple cider, white, and rice vinegar each bring unique flavors to dishes. Beyond its culinary uses, vinegar has an impressive range of applications, including cleaning, deodorizing, and removing stains due to its acidic nature.

Apple cider vinegar, in particular, is often praised for its potential health benefits, including aiding digestion, regulating blood sugar levels, and promoting skin health, though scientific evidence varies.

Vinegar also plays a role in preserving food, especially in pickling, as its acidity inhibits bacterial growth. Its simplicity, versatility, and affordability make it indispensable in kitchens and households worldwide. Whether for flavor or function, vinegar remains a timeless and essential product.

The experiment has two parts (Assuming that vinegar is only an aqueous solution of acetic acid).

Part 1: Determine the pH of the given vinegar sample. Using the pH value, calculate the concentration of acetic acid in the vinegar. Note that the dissociation constant (K_a) of acetic acid is 1.8×10^{-5} as obtained in the previous experiment.

Part 2: Find out the percentage of acetic acid present in the given vinegar solution provided to you using acid-base titration with a phenolphthalein indicator.

Dissociation equation for acetic acid in water



The expression for the dissociation constant (K_a)

Based on the dissociation equation, the expression for K_a is:

$$K_a = \frac{[\text{H}^+][\text{CH}_3\text{CO}_2^-]}{[\text{CH}_3\text{CO}_2\text{H}]}$$

You are given the value of $K_a = 1.8 \times 10^{-5}$.

Calculate the hydrogen ion concentration $[\text{H}^+]$ from pH

Assume that pH of the solution is 3. Using the pH formula, we can calculate the concentration of hydrogen ions $[\text{H}^+]$:

$$\text{pH} = -\log[\text{H}^+]$$

$$3.0 = -\log[\text{H}^+]$$

To find $[\text{H}^+]$, we can take the inverse logarithm of both sides:

$$[\text{H}^+] = 10^{-3.0}$$

$$[\text{H}^+] = 0.001(\text{M})$$

Now substitute the values in K_a expression

Since we assume that vinegar is only an aqueous solution of acetic acid, let the initial concentration of acetic acid be 'c'. After the dissociation, we will have 'c-x' concentration of CH_3COOH , 'x' concentration of H^+ ions and 'x' concentration of CH_3COO^- ions.

Now we can solve the K_a equation for the concentration of acetic acid, c :

$$1.8 \times 10^{-5} = \frac{0.001x}{c - x}$$

We know that $x \ll 0.001$, so we can write:

$$1.8 \times 10^{-5} \approx \frac{0.001x}{c}$$
$$c \approx \frac{0.001x}{1.8 \times 10^{-5}} \qquad c \approx \frac{0.001(0.001)}{1.8 \times 10^{-5}}$$

So, the concentration of acetic acid in the vinegar sample is approximately 0.0556M.

Reference: <https://www.vaia.com/en-us/textbooks/chemistry/chemistry-an-atoms-first-approach-2-edition/chapter-13/problem-80-a-typical-sample-of-vinegar-has-a-ph-of-30-assumi/>

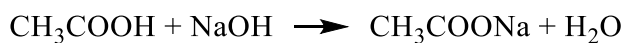
Titration of Vinegar

Vinegar is essentially a solution of acetic acid in water. The concentration of acetic acid in vinegar may be expressed as a molarity (in mol/L) or in or as a mass percent.

$$\text{Molarity} = \frac{\text{Moles of Acetic Acid}}{\text{Volume of Vinegar (in L)}}$$

$$\text{Mass \%} = \left(\frac{\text{Mass of Acetic Acid}}{\text{Mass of Vinegar}} \right) \times 100\%$$

In this experiment, a technique called titration will be used to measure the concentration of acetic acid in vinegar. Titration involves a controlled reaction between a solution with **a known concentration (the titrant)** and a solution with an **unknown concentration (the analyte)**. In this case, the titrant is an approximately 0.1 M aqueous solution of sodium hydroxide (NaOH), and the analyte is vinegar. When these are combined, a neutralization reaction occurs between the sodium hydroxide and the acetic acid present in the vinegar.



Sodium hydroxide will be added slowly to the vinegar in small increments from a burette, which is a device that allows precise delivery of specific volumes of solution. The NaOH will be added to the vinegar sample until all the acetic acid in the vinegar has fully reacted. This point marks the completion of the reaction and is known as the equivalence point of the titration.

To identify when the equivalence point is reached, a pH-sensitive indicator solution called phenolphthalein is added to the vinegar at the start of the titration. Phenolphthalein is colorless in acidic solutions like vinegar and turns deep pink in basic solutions like sodium hydroxide. At the equivalence point, a single drop of NaOH will turn the entire solution in the Erlenmeyer flask from colorless to pink (faint).

During the titration, the following data will be recorded:

- The molarity of the NaOH solution used
- The volume of NaOH solution needed to neutralize the vinegar
- The volume of vinegar used

Materials and Equipment

50-mL burette*, 5-mL volumetric pipette*, pipette bulb*, approximately 0.1 M NaOH (aqueous), vinegar, phenolphthalein indicator, burette stand, two Erlenmeyer flasks (250 mL or 125 mL), wash bottle filled with distilled water, funnel.

Safety

Take extra caution when handling sodium hydroxide (NaOH), as it is corrosive and can cause chemical burns to the skin. If any NaOH comes into contact with your skin, rinse immediately under running and inform your instructor of the incident.

Molarity of Acetic Acid in Vinegar

1. First, using the known molarity of the NaOH solution and the volume of NaOH required to reach the equivalence point, determine the moles of NaOH used in the titration.
2. Next, use this mole value of NaOH to find the moles of acetic acid (CH_3COOH) in the vinegar sample, based on the mole-to-mole ratio in the balanced equation.

3. Finally, calculate the molarity of acetic acid in the vinegar by using the moles of CH_3COOH and the volume of the vinegar sample used.

Mass Percent of Acetic Acid in Vinegar

1. First, convert the moles of acetic acid (CH_3COOH) in the vinegar sample (calculated previously) to mass using its molar mass.
2. Then, find the total mass of the vinegar sample by using the vinegar volume and assuming a density of 1.000 g/mL (the density of water).
3. Finally, calculate the mass percent of acetic acid in vinegar from the mass of CH_3COOH and the total mass of the vinegar sample.

Reference:

[https://chem.libretexts.org/Ancillary_Materials/Laboratory_Experiments/Wet_Lab_Experiments/General_Chemistry_Labs/Online_Chemistry_Lab_Manual/Chem_10_Experiments/11%3A_A_Titration_of_Vinegar_\(Experiment\)](https://chem.libretexts.org/Ancillary_Materials/Laboratory_Experiments/Wet_Lab_Experiments/General_Chemistry_Labs/Online_Chemistry_Lab_Manual/Chem_10_Experiments/11%3A_A_Titration_of_Vinegar_(Experiment))